

Lab 02: Network Security Groups and Application Security Groups

Student lab manual

Lab scenario

You have been asked to implement your organization's virtual networking infrastructure and test to ensure it is working correctly. In particular:

- The organization has two groups of servers: Web Servers and Management Servers.
- Each group of servers should be in its own Application Security Group.
- You should be able to RDP into the Management Servers, but not the Web Servers.
- The Web Servers should display the IIS web page when accessed from the internet.
- Network security group rules should be used to control network access.

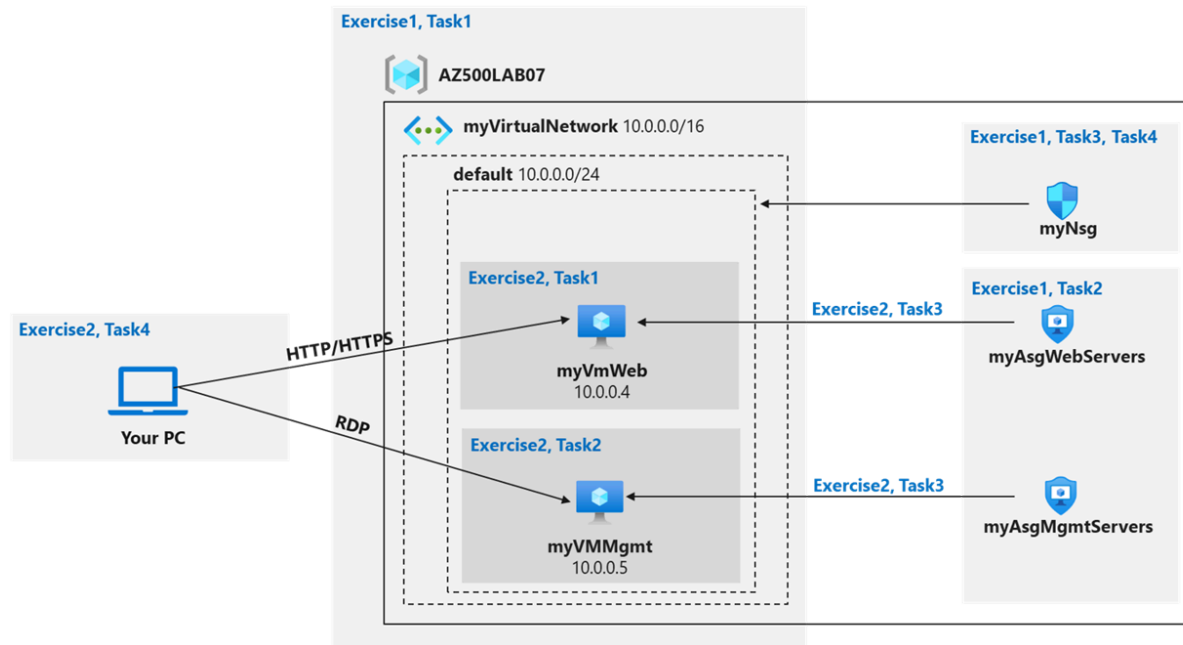
For all the resources in this lab, we are using the **East US** region. Verify with your instructor this is the region to use for class.

Lab objectives

In this lab, you will complete the following exercises:

- Exercise 1: Create the virtual networking infrastructure
- Exercise 2: Deploy virtual machines and test the network filters

Network and Application Security Groups diagram



Instructions

Exercise 1: Create the virtual networking infrastructure

Estimated timing: 20 minutes

For all the resources in this lab, we are using the **East (US)** region. Verify with your instructor this is the region to use for your class.

In this exercise, you will complete the following tasks:

- Task 1: Create a virtual network with one subnet.
- Task 2: Create two application security groups.
- Task 3: Create a network security group and associate it with the virtual network subnet.
- Task 4: Create inbound NSG security rules to all traffic to web servers and RDP to the management servers.

Task 1: Create a virtual network

In this task, you will create a virtual network to use with the network and application security groups.

1. Sign-in to the Azure portal <https://portal.azure.com/>.
Sign in to the Azure portal using an account that has the Owner or Contributor role in the Azure subscription you are using for this lab. In this **Cloudslice** lab, this account is LabUser-63327256@LODSPRODMSLEARNMCA.onmicrosoft.com
2. In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type Virtual networks and press the **Enter** key.
3. On the **Virtual networks** blade, click **+ Create**.
4. On the **Basics** tab of the **Create virtual network** blade, specify the following settings (leave others with their default values) and click **Next: IP Addresses**:

Setting	Value
Subscription	Name of the Azure subscription you are using in this lab
Resource group	Use the provided Resource Group named AZ500LAB07
Name	myVirtualNetwork
Region	East US

5. On the **IP addresses** tab of the **Create virtual network** blade, set the **IPv4 address space** to **10.0.0.0/16**, and, if needed, in the **Subnet name** column, click **default**, on the **Edit subnet** blade, specify the following settings and click **Save**:

Setting	Value
Subnet name	default
Subnet address range	10.0.0.0/24

6. Back on the **IP addresses** tab of the **Create virtual network** screen, click **Review + create**.
7. On the **Review + create** tab of the **Create virtual network** screen, click **Create**.

Task 2: Create application security groups

In this task, you will create an application security group.

1. In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type Application security groups and press the **Enter** key.
2. On the **Application security groups** blade, click **+ Create**.
3. On the **Basics** tab of the **Create an application security group** blade, specify the following settings:

Setting	Value
Resource group	AZ500LAB07
Name	myAsgWebServers
Region	East US

4. This group will be for the web servers.
5. Click **Review + create** and then click **Create**.
6. Navigate back to the **Application security groups** blade and click **+ Create**.
7. On the **Basics** tab of the **Create an application security group** blade, specify the following settings:

Setting	Value
Resource group	AZ500LAB07
Name	myAsgMgmtServers
Region	East US

8. This group will be for the management servers.
9. Click **Review + create** and then click **Create**.

Task 3: Create a network security group and associate the NSG to the subnet

In this task, you will create a network security group.

1. In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type Network security groups and press the **Enter** key.
2. On the **Network security groups** blade, click **+ Create**.
3. On the **Basics** tab of the **Create network security group** blade, specify the following settings:

Setting	Value
Subscription	Name of the Azure subscription you are using in this lab
Resource group	AZ500LAB07
Name	myNsg
Region	East US

4. Click **Review + create** and then click **Create**.
5. In the Azure portal, navigate back to the **Network security groups** blade and select the **myNsg** entry. Or select **Go to resource** if available.
6. On the **myNsg** blade, in the **Settings** section, click **Subnets** and then select **+ Associate**.
7. On the **Associate subnet** blade, specify the following settings and select **OK**:

Setting	Value
Virtual network	myVirtualNetwork
Subnet	default

Task 4: Create inbound NSG security rules to all traffic to web servers and RDP to the servers.

1. On the **myNsg** blade, in the **Settings** section, click **Inbound security rules**.
2. Review the default inbound security rules and then click **+ Add**.
3. On the **Add inbound security rule** blade, specify the following settings to allow TCP ports 80 and 443 to the **myAsgWebServers** application security group (leave all other values with their default values):

Setting	Value
Source	Any
Source port ranges	*
Destination	in the drop-down list, select Application security group and then click myAsgWebServers
Service	Custom
Destination port ranges	80,443
Protocol	TCP
Action	Allow
Priority	100
Name	Allow-Web-All

4. Select the **Add** button on the **Add inbound security rule** page, to create the new inbound rule.
5. On the **myNsg** blade, in the **Settings** section, click **Inbound security rules**, and then click **+ Add**.
6. On the **Add inbound security rule** blade, specify the following settings to allow the RDP port (TCP 3389) to the **myAsgMgmtServers** application security group (leave all other values with their default values):

Setting	Value
Source	Any
Source port ranges	*

Destination	in the drop-down list, select Application security group and then click myAsgMgmtServers
Service	Custom
Destination port ranges	3389
Protocol	TCP
Action	Allow
Priority	110
Name	Allow-RDP-All

7. Select **Add** on the **Add inbound security rule** page, to create the new inbound rule.

Result: You have deployed a virtual network, network security with inbound security rules, and two application security groups.

Exercise 2: Deploy virtual machines and test network filters

Estimated timing: 25 minutes

In this exercise, you will complete the following tasks:

- Task 1: Create a virtual machine to use as a web server.
- Task 2: Create a virtual machine to use as a management server.
- Task 3: Associate each virtual machines network interface to it's application security group.
- Task 4: Test the network traffic filtering.

Task 1: Create a virtual machine to use as a web server

In this task, you will create a virtual machine to use as a web server.

1. In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type Virtual machines and press the **Enter** key.

2. On the **Virtual machines** blade, click **+ Create** and, in the dropdown list, click **Virtual machine**.
3. On the **Basics** tab of the **Create a virtual machine** blade, specify the following settings (leave others with their default values):

Setting	Value
Subscription	the name of the Azure subscription you will be using in this lab
Resource group	AZ500LAB07
Virtual machine name	myVmWeb
Region	(US)East US
Availability options	No infrastructure redundancy required
Security type	Standard
Image	Windows Server 2022 Datacenter: Azure Edition- x64 Gen2
Size	Standard D2s v3
Username	Student
Password	Please create your own password and record it for future reference in subsequent labs
Confirm password	Retype your password
Public inbound ports	None
Would you like to use an existing Windows Server License	No

4. For public inbound ports, we will rely on the precreated NSG.
5. Click **Next: Disks >** and, on the **Disks** tab of the **Create a virtual machine** blade, set the **OS disk type** to **Standard HDD** and click **Next: Networking >**.
6. On the **Networking** tab of the **Create a virtual machine** blade, select the previously created network **myVirtualNetwork** and the **default (10.0.0.0/24)** subnet.
7. Under **NIC network security group** select **None**.

8. Click **Next: Management >**, then click **Next: Monitoring >**. On the **Monitoring** tab of the **Create a virtual machine** blade, verify the following setting:

Setting	Value
Boot diagnostics	Enabled with managed storage account (recommended)

9. Click **Review + create**, on the **Review + create** blade, ensure that validation was successful and click **Create**.

Task 2: Create a virtual machine to use as a management server.

In this task, you will create a virtual machine to use as a management server.

1. In the Azure portal, navigate back to the **Virtual machines** blade, click **+ Create**, and, in the dropdown list, click **Virtual machine**.
2. On the **Basics** tab of the **Create a virtual machine** blade, specify the following settings (leave others with their default values):

Setting	Value
Subscription	the name of the Azure subscription you will be using in this lab
Resource group	AZ500LAB07
Virtual machine name	myVMMgmt
Region	(US)East US
Availability options	No infrastructure redundancy required
Security type	Standard
Image	Windows Server 2022 Datacenter: Azure Edition - x64 Gen2
Size	Standard D2s v3
Username	Student
Password	Please use your personal password created in Lab 02 > Exercise 2 > Task 1 > Step 3.
Public inbound ports	None

Already have a Windows Server license	No
---------------------------------------	----

3. For public inbound ports, we will rely on the precreated NSG.
4. Click **Next: Disks >** and, on the **Disks** tab of the **Create a virtual machine** blade, set the **OS disk type** to **Standard HDD** and click **Next: Networking >**.
5. On the **Networking** tab of the **Create a virtual machine** blade, select the previously created network **myVirtualNetwork** and the **default (10.0.0.0/24)** subnet.
6. Under **NIC network security group** select **None**.
7. Click **Next: Management >**, then click **Next: Monitoring >**. On the **Monitoring** tab of the **Create a virtual machine** blade, verify the following setting:

Setting	Value
Boot diagnostics	Enabled with managed storage account (recommended)

8. Click **Review + create**, on the **Review + create** blade, ensure that validation was successful and click **Create**.
Wait for both virtual machines to be provisioned before continuing.

Task 3: Associate each virtual machine's network interface to its application security group.

In this task, you will associate each virtual machines network interface with the corresponding application security group. The myVMWeb virtual machine interface will be associated to the myAsgWebServers ASG. The myVMMgmt virtual machine interface will be associated to the myAsgMgmtServers ASG.

1. In the Azure portal, navigate back to the **Virtual machines** blade and verify that both virtual machines are listed with the **Running** status.
2. In the list of virtual machines, click the **myVMWeb** entry.
3. On the **myVMWeb** blade, in the **Networking** section, click **Network settings** and then, on the **myVMWeb | Networking settings** blade, click the **Application security groups** tab.
4. Click **+ Add application security groups**, in the **Application security group** list, select **myAsgWebServers**, and then click **Save**.
5. Navigate back to the **Virtual machines** blade and in the list of virtual machines, click the **myVMMgmt** entry.

6. On the **myVMMgmt** blade, in the **Networking** section, click **Networking settings** and then, on the **myVMMgmt | Networking settings** blade, click the **Application security groups** tab.
7. Click + **Add application security groups**, in the **Application security group** list, select **myAsgMgmtServers**, and then click **Add**.

Task 4: Test the network traffic filtering

In this task, you will test the network traffic filters. You should be able to RDP into the myVMMgmt virtual machine. You should be able to connect from the internet to the myVMWeb virtual machine and view the default IIS web page.

1. Navigate back to the **myVMMgmt** virtual machine blade.
2. On the **myVMMgmt** Overview blade, click **Connect** and, in the drop down menu, click **Connect**.
3. Download the RDP file and use it to connect to the **myVMMgmt** Azure VM via Remote Desktop. When prompted to authenticate, provide the following credentials:

Setting	Value
User name	Student
Password	Please use your personal password created in Lab 02 > Exercise 1 > Task 1 > Step 9.

4. Verify that the Remote Desktop connection was successful. At this point you have confirmed you can connect via Remote Desktop to myVMMgmt.
5. In the Azure portal, navigate to the **myVMWeb** virtual machine blade.
6. On the **myVMWeb** blade, in the **Operations** section, click **Run command** and then click **RunPowerShellScript**.
7. On the **Run Command Script** pane, run the following to install the Web server role on **myVmWeb**:
8.

```
Install-WindowsFeature -name Web-Server  
-IncludeManagementTools
```
9. Wait for the installation to complete. This might take a couple of minutes. At that point, you can verify that myVMWeb can be accessed via HTTP/HTTPS.
10. In the Azure portal, navigate back to the **myVMWeb** blade.
11. On the **myVMWeb** blade, identify the **Public IP address** of the myVmWeb Azure VM.
12. Open another browser tab and navigate to IP address you identified in the previous step.
The browser page should display the default IIS welcome page because port 80 is allowed inbound from the internet based on the setting of the **myAsgWebServers**

application security group. The network interface of the myVMWeb Azure VM is associated with that application security group.

Result: You have validated that the NSG and ASG configuration is working and traffic is being correctly managed.

Clean up resources

Remember to remove any newly created Azure resources that you no longer use. Removing unused resources ensures you will not incur unexpected costs.

1. Open the Cloud Shell by clicking the first icon in the top right of the Azure Portal. If prompted, select **PowerShell** and **No storage account required**, select the name of your subscription, and then select **Apply**.
2. Ensure **PowerShell** is selected in the drop-down menu in the upper-left corner of the Cloud Shell pane.
3. In the PowerShell session within the Cloud Shell pane, run the following to remove the resource group you created in this lab:
4.

```
Remove-AzResourceGroup -Name "AZ500LAB07" -Force -AsJob
```
5. Close the **Cloud Shell** pane.

Screenshots

Microsoft Azure

Q Search resources, services, and docs (G+ /)

Home > Network foundation | Virtual networks

Create virtual network ...

Basics

Security

Address space

Tags

Review + create

[Learn more.](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

MOC Subscription-lod52875296

Resource group *

AZ500LAB07

[Create new](#)

Instance details

Virtual network name *

myVirtualNetwork

Region * ⓘ

(US) East US

[Deploy to an Azure Extended Zone](#)

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Review + create

Create virtual network ...

Basics Security Address space Tags Review + create

10.0.0.0/16

Delete address space

Starting address

Size

10.0.0.0

/16

10.0.0.0 - 10.0.255.255

65,536 addresses

Subnets	IP address range	Size	NAT gateway
default	10.0.0.0 - 10.0.0.255	/24 (256 addresses)	-

Add IPv4 address space |

Advertised address prefixes

Advertised gateway prefixes let you control route advertisements by specifying which IP address ranges Azure gateways advertise to on premises networks, instead of advertising all virtual network and spoke CIDRs.

+ Add IPv4 address prefix + Add IPv6 address prefix



Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template



Your deployment is complete



Deployment name : myVirtualNetwork-1783567359958
Subscription : [MOC Subscription-Iod52875296](#)
Resource group : [AZ500LAB07](#)

Start time : 7/8/2026, 8:23:07 PM
Correlation ID : bb10be2d-3bf2-425b-afc2-68bec310ade9

> Deployment details

< Next steps

Go to resource

Create an application security group ...

Basics Tags Review + create

Project details

Subscription *

Resource group *

[Create new](#)

Instance details

Name *

Region *

Review + create

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CreateApplicationSecurityGroupBladeViewModel | Overview ...

Deployment

× << [Delete](#) [Cancel](#) [Redeploy](#) [Download](#) [Refresh](#)

Overview

[Inputs](#)

[Outputs](#)

[Template](#)

✓ Your deployment is complete

Deployment name : CreateApplicationSecurityGroupBladeViewM... Start time : 7/8/2026, 8:24:21 PM
Subscription : [MOC Subscription-lod52875296](#) Correlation ID : 33b5bdc9-62d7-4512-ad15-ef3f47efa46c
Resource group : [AZ500LAB07](#)

> Deployment details

✓ Next steps

[Go to resource](#)

Create an application security group ...

Basics Tags Review + create

Project details

Subscription *	MOC Subscription-lod52875296
Resource group *	AZ500LAB07

[Create new](#)

Instance details

Name *	myAsgMgmtServers
Region *	East US

Review + create

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CreateApplicationSecurityGroupBladeViewModel | Overview ...

Deployment

Search × << Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template



Your deployment is complete

Deployment name : CreateApplicationSecurityGroupBladeViewM... Start time : 7/8/2026, 8:25:20 PM
Subscription : MOC Subscription-lod52875296 Correlation ID : c4c6c0fe-d337-483f-9e41-f2a7abaf925f
Resource group : AZ500LAB07

> Deployment details

✓ Next steps

[Go to resource](#)

Create network security group ...

Basics Tags Review + create

Project details

Subscription *

MOC Subscription-lod52875296

Resource group *

AZ500LAB07

Create new

Instance details

Name *

myNsg

Region *

East US

Review + create < Previous Next : Tags > [Download a template for automation](#)

CreateNetworkSecurityGroupBladeV2-20260708202558 | Overview ...

Search Delete Cancel Redeploy Download Refresh

Overview

- Inputs
- Outputs
- Template

✔ Your deployment is complete

Deployment name : CreateNetworkSecurityGroupBladeV2-20260... Start time : 7/8/2026, 8:26:36 PM
Subscription : MOC Subscription-lod52875296 Correlation ID : 6de0c66a-d751-434d-8a3c-3359fa7644f9
Resource group : AZ500LAB07

> Deployment details

✓ Next steps

Go to resource

myNsg | Subnets

Network security group

[+ Associate](#)

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Inbound security rules

Outbound security rules

Network interfaces

Subnets

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Name	Address range
------	---------------

No results.

Associate subnet

myNsg

Virtual network

myVirtualNetwork (AZ500LAB07)

Subnet *

default

[OK](#)

Add or remove favorites by pressing Ctrl+Shift+F

myNsg | Inbound security rules

Network security group

[+ Add](#) [Hide default rules](#) [Refresh](#) [Delete](#) [Give feedback](#)

Overview

Activity log

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Network security group security rules are evaluated by priority using the combination of source, source port ranges, destination, destination port ranges, protocol, and direction. Rules with the same priority and direction can't have the same source, source port ranges, destination, destination port ranges, protocol, and direction. You can't delete default security rules.

Port == all

Protocol == all

Priority	Name	Port	Protocol
☐ 65000	AllowVnetInBound	Any	Any
☐ 65001	AllowAzureLoadBalanc...	Any	Any
☐ 65500	DenyAllInBound	Any	Any

Add inbound security rule

myNsg

Source

Any

Source port ranges *

*

Destination

Application security group

Destination application security groups

myAsgWebServers

Filter the application security groups

Service

Custom

Destination port ranges *

80,443

Protocol

☐ Any☒ TCP☐ UDP☐ ICMPv4☐ ICMPv6[Add](#)[Cancel](#)[Give feedback](#)

Add or remove favorites by pressing Ctrl+Shift+F



Add inbound security rule



myNsg

Custom



Destination port ranges * ⓘ

80,443



Protocol



Any



TCP



UDP



ICMPv4



ICMPv6

Action



Allow



Deny

Priority * ⓘ

100

Name *

Allow-Web-All



Description

Add

Cancel

 Give feedback

Home > CreateNetworkSecurityGroupBladeV2-20260708202558 | Overview > myNsg

myNsg | Inbound security rules

Network security group

Search

◦ << + Add Hide default rules Refresh Delete Give feedback

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Help

Network security group security rules are evaluated by priority using the combination of source, source port, destination, destination port, and protocol. Rules can't have the same priority and direction as an existing rule. You can't delete default security rules.

Filter by name

Port == all

Protocol == all

Priority	Name	Port	Protocol
☐ 100	Allow-Web-All	80,443	TCP
☐ 65000	AllowVnetInBound	Any	Any
☐ 65001	AllowAzureLoadBalanc...	Any	Any
☐ 65500	DenyAllInBound	Any	Any

Add inbound security rule

myNsg

Source

Any

Source port ranges

*

Destination

Application security group

Destination application security groups

myAsgMgmtServers

Filter the application security groups

Service

Custom

Destination port ranges

3389

Protocol

Any

TCP

UDP

ICMPv4

ICMPv6

Add

Cancel

Give feedback

Add or remove favorites by pressing Ctrl+L+Shift+F



Add inbound security rule



myNsg

Custom



Destination port ranges * ⓘ

3389



Protocol



Any



TCP



UDP



ICMPv4



ICMPv6

Action



Allow



Deny

Priority * ⓘ

110

Name *

Allow-RDP-All



Description

Home > Compute infrastructure

Compute infrastructure | Virtual machines

Get workload templates for VMs | List all alerts for virtual machines | Analyze alerts across virtual machines

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Overview

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Virtual Machine Scale Set (VMSS)

Compute Fleet

Arc Machines

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> Capacity + placement

> Related services

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> Help

+ Create | Reservations | Manage view | Refresh | Export to CSV | Open query | Assign tags | Start | Restart | Group by none

Type equals all | Resource Group equals all | Location equals all | Add filter

Virtual machine

Best for lower-traffic workloads, testing, or to control or highly customize apps, OS, or file system. If your workload or traffic starts to grow, a VM can later be attached to a Virtual Machine Scale Set (VMSS).

Virtual machine scale set (VMSS)

Built-in scaling, performance optimization, load balancing, and batch management for 1 to 1,000 VMs (no added cost). Include multiple VM sizes, zones, regions, and domains, along with discounted Spot VMs.

Hybrid, preconfigured, and high volume solutions

Explore pre-configured Starter kits for Linux and Windows, Azure Arc hybrid infrastructure solutions, and more.

No virtual machines to display

machine that runs Linux or Windows. Select an image from the marketplace or use your own customized image.

+ Create

[Learn more about Windows virtual machines](#)

[Learn more about Linux virtual machines](#)

Showing 1 - 0 of 0. Display count: auto

[Give feedback](#)

https://portal.azure.com/#blade/Microsoft_Azure_Compute/createUiDef/%7B%22isFullCreate%22%3Atrue%7D

Create a virtual machine

Help me create a VM optimized for high availability

Help me create a low cost VM

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

MOC Subscription-lod52875296

Resource group * ⓘ

AZ500LAB07

[Create new](#)

Instance details

Virtual machine name * ⓘ

myVmWeb

Region * ⓘ

(US) East US

[Deploy to an Azure Extended Zone](#)

Availability options ⓘ

No infrastructure redundancy required

Security type ⓘ

Standard

Image * ⓘ

 Windows Server 2022 Datacenter: Azure Edition - x64 Gen2

[See all images](#) | [Configure VM generation](#)

< Previous

Next : Disks >

Review + create

Create a virtual machine



Help me create a VM optimized for high availability

Help me create a low cost VM

VM architecture ⓘ

☐ Arm64

☒ x64

i Arm64 is not supported with the selected image.

Run with Azure Spot discount ⓘ

☐

Size * ⓘ

Standard_D2s_v3 - 2 vcpus, 8 GiB memory (\$137.24) ✓

[See all sizes](#) | [Customize cores](#)

Enable Hibernation ⓘ

☐

i Hibernate is not supported by the size that you have selected. Choose a size that is compatible with Hibernation to enable this feature. [Learn more](#) ↗

Administrator account

Username * ⓘ

Student ✓

Password *

..... ✓

Confirm password *

..... ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ

☒ None

☐ Allow selected ports

< Previous

Next : Disks >

Review + create

Create a virtual machine

...



Help me create a VM optimized for high availability

Help me create a low cost VM

Username * ⓘ

Student



Password *

.....



Confirm password *

.....



Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ

☒ None

☐ Allow selected ports

Select inbound ports

Select one or more ports



All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

Licensing

Save up to 49% with a license you already own using Azure Hybrid Benefit. [Learn more](#) ↗

Would you like to use an existing
Windows Server license? ⓘ

☐

[Review Azure hybrid benefit compliance](#) ↗

< Previous

Next : Disks >

Review + create

Create a virtual machine



Help me create a VM optimized for high availability

Help me create a low cost VM

i There is a charge for the underlying storage resources consumed by your virtual machine. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host **i**

☐

i Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size **i**

Image default (127 GiB) **v**

OS disk type * **i**

Standard HDD (locally-redundant storage) **v**

The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

Delete with VM **i**

☒

Key management **i**

Platform-managed key **v**

Enable Ultra Disk compatibility **i**

☐

Ultra disk is supported in Availability Zone(s) 1,2,3 for the selected VM size Standard_D2s_v3.

Data disks for myVmWeb

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Next : Networking >

Review + create

Create a virtual machine

...

Help me create a VM optimized for high availability

Help me create a low c

Basics

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Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.

[Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network *

myVirtualNetwork

Create new

Subnet *

default (10.0.0.0/24)

Manage subnet configuration

Public IP

(new) myVmWeb-ip

Create new

Public IP addresses have a nominal charge. [Estimate price](#)

NIC network security group

☒ None

☐ Basic

☐ Advanced

The selected subnet 'default (10.0.0.0/24)' is already associated to a network security group 'myNsg'. We recommend managing connectivity to this virtual machine via the existing network security group instead of creating a new one here.

< Previous

Next : Management >

Review + create

[Basics](#)[Disks](#)[Networking](#)[Management](#)[Monitoring](#)[Advanced](#)[Tags](#)[Review + create](#)

Configure monitoring options for your VM.

Alerts

Enable recommended alert rules ⓘ

☐

Diagnostics

Boot diagnostics ⓘ

☒

Enable with managed storage account (recommended)

☐

Enable with custom storage account

☐

Disable



There is a nominal charge for the managed storage account.

Enable OS guest diagnostics ⓘ

☐

Health

Enable application health monitoring ⓘ

☐[< Previous](#)[Next : Advanced >](#)[Review + create](#)

CreateVm-microsoftwindowsserver.windowsserver2022-20260708203913 | Overview

Search x << Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : CreateVm-microsoftwindowsserver.windowss... Start time : 7/8/2026, 8:43:30 PM
Subscription : MOC Subscription-lod52875296 Correlation ID : e6986421-8134-459c-a21c-dc909f7d1c2b
Resource group : AZ500LAB07

> Deployment details

✓ Next steps

[Set up auto-shutdown](#) Recommended

[Monitor VM health, performance, and network dependencies](#) Recommended

[Run a script inside the virtual machine](#) Recommended

[Go to resource](#)

[Create another VM](#)

[Scale out your VM](#)

Create a virtual machine



[Help me create a VM optimized for high availability](#)

[Help me choose the r](#)

Basics

Disks

Networking

Management

Monitoring

Advanced

Tags

Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

MOC Subscription-lod52875296

Resource group * ⓘ

AZ500LAB07

[Create new](#)

Instance details

Virtual machine name * ⓘ

myVMMgmt

Region * ⓘ

(US) East US

[Deploy to an Azure Extended Zone](#)

Availability options ⓘ

No infrastructure redundancy required

Security type ⓘ

Standard

Image * ⓘ

Windows Server 2022 Datacenter: Azure Edition - x64 Gen2

< Previous

Next : Disks >

Review + create

Create a virtual machine

...



Help me create a VM optimized for high availability

Help me choose the r

VM architecture ⓘ

- ☐ Arm64
- ☒ x64

i Arm64 is not supported with the selected image.

Run with Azure Spot discount ⓘ

☐

Size * ⓘ

Standard_D2s_v3 - 2 vcpus, 8 GiB memory (\$137.24) ▼

[See all sizes](#) | [Customize cores](#)

Enable Hibernation ⓘ

☐

i Hibernation is not supported by the size that you have selected. Choose a size that is compatible with Hibernation to enable this feature. [Learn more](#) ↗

Administrator account

Username * ⓘ

Student ✓

Password *

..... ✓

Confirm password *

..... ✓

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ

- ☒ None

< Previous

Next : Disks >

Review + create

Create a virtual machine

[Help me create a VM optimized for high availability](#)[Help me choose the right VM](#)

Username * ⓘ

Student



Password *

.....



Confirm password *

.....



Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ



None



Allow selected ports

Select inbound ports

Select one or more ports



All traffic from the internet will be blocked by default. You will be able to change inbound port rules in the VM > Networking page.

Licensing

Save up to 49% with a license you already own using Azure Hybrid Benefit. [Learn more](#)

Would you like to use an existing Windows Server license? ⓘ

☐

[Review Azure hybrid benefit compliance](#)

< Previous

Next : Disks >

Review + create

Create a virtual machine

...



Help me create a VM optimized for high availability

Help me choose the r

Basics

Disks

Networking

Management

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Tags

Review + create

i Standard HDD OS disks will be retired on September 8, 2028. For new OS disks, use Standard SSD or Premium SSD instead. Migrate existing Standard HDD OS disks to Standard SSD or Premium SSD. [Click to learn more](#)

×

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

i There is a charge for the underlying storage resources consumed by your virtual machine. [Learn more](#)

VM disk encryption

Azure disk storage encryption automatically encrypts your data stored on Azure managed disks (OS and data disks) at rest by default when persisting it to the cloud.

Encryption at host **i**

☐

i Encryption at host is not registered for the selected subscription. [Learn more](#)

OS disk

OS disk size **i**

Image default (127 GiB)

▼

OS disk type * **i**

Standard HDD (locally-redundant storage)

▼

The selected VM size supports premium disks. We recommend Premium SSD for high IOPS workloads. Virtual machines with Premium SSD disks qualify for the 99.9% connectivity SLA.

< Previous

Next : Networking >

Review + create

Create a virtual machine

...



Help me create a VM optimized for high availability

Help me choose the ri

Basics Disks **Networking** Management Monitoring Advanced Tags Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. [Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * ⓘ

myVirtualNetwork

[Create new](#)

Subnet * ⓘ

default (10.0.0.0/24)

[Manage subnet configuration](#)

Public IP ⓘ

(new) myVMMgmt-ip

[Create new](#)

i Public IP addresses have a nominal charge. [Estimate price](#)

NIC network security group ⓘ

- ☒ None
☐ Basic
☐ Advanced

i The selected subnet 'default (10.0.0.0/24)' is already associated to a network security group 'myNsg'. We recommend managing connectivity to this virtual machine via the existing network security group instead of creating a new one here.

< Previous

Next : Management >

Review + create

Create a virtual machine

[Help me create a VM optimized for high availability](#)[Help me choose a VM size](#)

Basics Disks Networking Management **Monitoring** Advanced Tags Review + create

Configure monitoring options for your VM.

Alerts

Enable recommended alert rules ⓘ ☐

Diagnostics

Boot diagnostics ⓘ

- ☒ Enable with managed storage account (recommended)
- ☐ Enable with custom storage account
- ☐ Disable

i There is a nominal charge for the managed storage account.

Enable OS guest diagnostics ⓘ ☐

Health

Enable application health monitoring ⓘ ☐

[< Previous](#)[Next : Advanced >](#)[Review + create](#)



CreateVm-microsoftwindowsserver.windowsserver2022-20260708204459 | Overview ...

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

✓ Your deployment is complete

Deployment name : CreateVm-microsoftwindowsserver.windowss... Start time : 7/8/2026, 8:48:33 PM
Subscription : MOC Subscription-Iod52875296 Correlation ID : ff15eb80-3bc9-4ad6-8257-9440448c7409
Resource group : AZ500LAB07

> Deployment details

✓ Next steps

[Set up auto-shutdown](#) Recommended
[Monitor VM health, performance, and network dependencies](#) Recommended
[Run a script inside the virtual machine](#) Recommended

[Go to resource](#)

[Create another VM](#)

[Scale out your VM](#)

Home > myVmWeb



myVmWeb
Virtual machine

Application security groups ☆ ...

Search

+ Add application security groups Remove Refresh Give feedback

Network interface / IP configuration
myvmweb20 (primary) / ipconfig1 (primary)

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Add or remove favorites by pressing Ctrl+Shift+F

No application security groups

You can use application security groups to configure network security as natural extension of an application's structure, by arbitrarily grouping VMs and defining network security policies based on those groups.

[Add application security group](#)

[Learn more](#)

Add application security groups

Use application security groups (ASGs) to configure network security as natural extension of an application's structure, by arbitrarily grouping VMs and defining network security policies based on those groups.

Search ASGs

☐	Name	Resource group
☒	myAsgWebServers	AZ500LAB07
☐	myAsgMgmtServers	AZ500LAB07

[Add](#)

[Cancel](#)

[Give feedback](#)

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

LabUser-63353875@LO...
LODS-PROD-MSLEARN-MCA (ID...

Home > myVMMgmt

myVMMgmt Application security groups ☆ ...

Virtual machine

Search

+ Add application security groups Remove Refresh Give feedback

Network interface / IP configuration
myvmgmt325 (primary) / ipconfig1 (primary)

Overview
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Backup + disaster recovery

No application security groups

You can use application security groups to configure network security as a natural extension of an application's structure, by arbitrarily grouping VMs and defining network security policies based on those groups.

Add application security groups

Learn more

Add application security groups

Use application security groups (ASGs) to configure network security as a natural extension of an application's structure, by arbitrarily grouping VMs and defining network security policies based on those groups.

Search ASGs

Name	Resource group
myAsgWebServers	AZ500LAB07
myAsgMgmtServers	AZ500LAB07

Add Cancel Give feedback

myVmWeb | Connect ☆ ...

Virtual machine

Search

Refresh Reset password or keys Manage JIT Troubleshoot Feedback

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Diagnose and solve problems
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Bastion
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Load balancing
Application security groups
Network manager
Settings
Availability + scale
Security

Now view, configure, and even save your connection settings — all in one place. Have comments or suggestions for our new Connect experience? [Provide feedback](#)

Native RDP

MOST POPULAR LOCAL MACHINE

Source machine

Source machine OS Windows

Source IP address Local IP | 103.18.87.247 [Connecting over a VPN?](#)

Destination VM

VM IP address Public IP | 20.106.246.65

VM port 3389

Connection prerequisites

VM access Port 3389 is not accessible from source IP(s) [View blocking NSG rules](#)

Check access

Connect using RDP file

Download and open file to connect

Download RDP file

Username Student [Forgot password? Reset password](#)

Edit settings

More ways to connect (3)

myVMMgmt | Connect

Virtual machine

Search

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Connect

Bastion

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Availability + scale

Security

Now view, configure, and even save your connection settings — all in one place. Have comments or suggestions for our new Connect

RefreshReset password or keysManage JITTroubleshootFeedback

Native RDP

MOST POPULARLOCAL MACHINE

Source machine

Source machine OSWindows

Source IP addressLocal IP | 103.18.87.247Connecting over a VPN?

Destination VM

VM IP addressPublic IP | 20.55.16.188

VM port3389

Connection prerequisites

VM accessPort 3389 is accessible from source IP(s) View applied NSG rules

Check access

Connect using RDP file

Download and open file to connectDownload RDP file

UsernameStudentForgot password? Reset password

Edit settings

More ways to connect (3)

Server Manager

20.55.16.188

Server Manager > Local Server

ManageToolsViewHelp

Dashboard

Local Server

All Servers

File and Storage Services

PROPERTIES

For myVMMgmt

Computer namemyVMMgmt

WorkgroupWORKGROUP

Last installed updatesToday at 3:09 AM

Windows UpdateInstall updates automatically using Windows Update

Last checked for updatesToday at 3:08 AM

Microsoft Defender FirewallPrivate: On

Remote managementEnabled

Remote DesktopEnabled

NIC TeamingDisabled

EthernetIPv4 address assigned by DHCP, IPv6 enabled

Azure Arc ManagementDisabled

Microsoft Defender AntivirusReal-Time Protection: On

Feedback & DiagnosticsSettings

IE Enhanced Security ConfigurationOn

Time zone(UTC) Coordinated Universal Time

Product ID00446-90000-00000-AA651 (activated)

Operating system versionMicrosoft Windows Server 2022 Datacenter Azure Edition

Hardware informationMicrosoft Corporation Virtual Machine

ProcessorsIntel(R) Xeon(R) Platinum 8370C CPU @ 2.80GHz

Installed memory (RAM)8 GB

Total disk space142.45 GB

EVENTS

All events | 4 total

Filter

Server NameIDSeveritySourceLogDate and Time

myVMMgmt10016WarningMicrosoft-Windows-DistributedCOMSystem7/10/2026 3:10:57 AM

myVMMgmt7031ErrorMicrosoft-Windows-Service Control ManagerSystem7/10/2026 3:06:12 AM

myVMMgmt10016WarningMicrosoft-Windows-DistributedCOMSystem7/10/2026 3:04:48 AM

myVMMgmt15WarningMicrosoft-Windows-WininitSystem7/10/2026 3:03:42 AM

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

LabUser-63353875@l.O...
LODS-PROD-MSLEARN-MCA (ID...

Home > myVmWeb

myVmWeb | Run command

Virtual machine

Search

Now it's possible to execute multiple scripts at t deployment. Learn more about the updated Run

Run Command uses the VM agent to let you run a machine and application maintenance. Select a con

Run command

Updates

Health monitoring

Configuration management

Policies

Inventory

Change tracking

Monitoring

Automation

Name

RunPowerShellScript

DisableNLA

DisableWindowsUpdate

EnableAdminAccount

EnableEMS

EnableRemotePS

EnableWindowsUpdate

FleetDiagnosticsWindows

IMDSCertCheck

IPConfig

RDPSettings

ResetRDPcert

SetRDPport

Run Command Script

RunPowerShellScript

PowerShell Script

1 Install-WindowsFeature -name Web-Server -IncludeManagementTools

Run

Run Command Script



RunPowerShellScript

 Script execution complete

PowerShell Script

```
1 Install-WindowsFeature -name Web-Server -IncludeManagementTools
```

Run

Output

Success	Restart Needed	Exit Code	Feature Result
True	No	Success	{Common HTTP Features, Default Document, D...

myVmWeb

Virtual machine

Help me copy this VM in any region

Manage this VM with Azure CLI

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Start

Restart

Stop

Hibernate

Capture

Delete

Refresh

Scale

Open in mobile

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Diagnose and solve problems

Monitor

Resource visualizer

Connect

Networking

Settings

Availability + scale

Security

Backup + disaster recovery

Operations

Auto-shutdown

Essentials

Resource group (move) : AZ500LAB07

Status : Running

Location : East US

Subscription (move) : MQC Subscription-Iod52879133

Subscription ID : ce6c06a0-902b-4e66-93d6-1f52180a395d

Operating system : Windows (Windows Server 2022 Datacenter Azure Edition)

Size : Standard D2s v3 (2 vcpus, 8 GiB memory)

Primary NIC public IP : 20.106.246.65

1 associated public IPs

Virtual network/subnet : myVirtualNetwork/default

DNS name : Not configured

Health state : -

Time created : 7/10/2026, 3:03 AM UTC

Tags (edit) : Add tags

PropertiesMonitoringCapabilities (8)RecommendationsTutorials

Virtual machine

Computer name : myVmWeb

Operating system : Windows (Windows Server 2022 Datacenter Azure Edition)

Networking

Public IP address : 20.106.246.65 (Network interface myvmweb653)

1 associated public IPs

Run Command Script - Microsoft

myVmWeb - Microsoft Azure

Home - Microsoft Azure

IIS Windows Server

Not secure 20.106.246.65

Update

Chat

Windows Server

Internet Information Services

Welcome

Bienvenue

Tervetuloa

ようこそ

Benvenuto

歡迎

Bem-vindo

Bienvenido

Hoş geldiniz

ברוכים הבאים

Welkom

Vítejte

Καλώς ορίσαστε

Välkommen

환영합니다

Добро пожаловать

Üdvözlünk

مرحبا

歡迎

Willkommen

Velkommen

Witamy

Microsoft